Comparative Evaluation of Real-Time Communication Protocols in a Real Vehicle Teleoperation Platform

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Abstract. This work presents a comparative evaluation of real-time communication protocols for the teleoperation of autonomous vehicles. Using a real vehicle and a fully operational teleoperation station, performance is tested under identical conditions. Metrics such as latency, jitter, and packet loss are measured to support the selection of safe and efficient teleoperation systems for real-world use.

Keywords: Teleoperation · Real-time communication · Video streaming

1 Introduction

Autonomous vehicles rely on sensors and AI-based perception to navigate, but their performance can encounter difficulties in extreme or unpredictable situations. Teleoperation can serve as an essential fallback, enabling human operators to remotely control vehicles through real-time video and command transmission.

Previous studies have examined communication technologies for teleoperation [1, 2], but most study only a small number of protocols or rely on simulation-based environments. This work advances the state of the art by comparing a large range of transport and streaming technologies under identical conditions using a real autonomous vehicle testbed, helping to better understand which protocols are most suitable for safe teleoperation.

2 Methodology

2.1 System Overview

As shown in Fig. 1, the experimental setup consists of a real vehicle equipped with drive-by-wire, connected over a 5G network to a teleoperation station with driving controls and a monitoring interface. Video is sent from the vehicle to the station, while control commands are transmitted back to close the loop.

2.2 Evaluated Protocols

Transport Baselines: UDP provides low-latency transmission and may experience packet loss, while TCP delivers data reliably at the cost of higher latency.

Streaming Protocols: RTP provides real-time streaming with jitter buffering, while SRT adds packet recovery to maintain low latency.

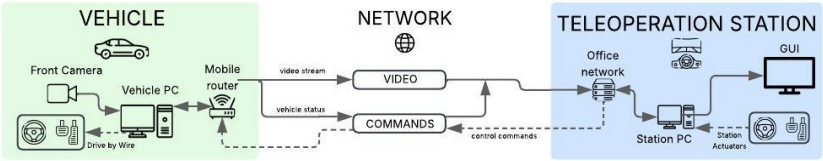

Fig. 1: System architecture with the vehicle, network, and teleoperation station.

Web-Based Protocols: WebSockets offer wide compatibility at the expense of latency, while WebRTC and WebTransport provide low-latency bidirectional communication.

3 Results and Conclusion

The tests were performed over a 5G network following a predefined route, covering three scenarios: video only, commands only, and simultaneous transmission. Latency, jitter, and packet loss were measured to compare protocol performance under real conditions.

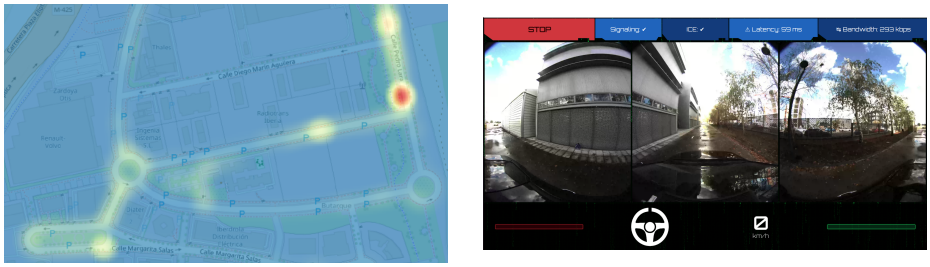

Fig. 2: Heatmap for WebRTC (left) showing areas of higher latency, alongside the teleoperation interface (right) displaying real-time video and commands.

The results highlight general differences in latency, reliability, and responsiveness across protocols. As an example, Fig. 2 shows a heatmap for WebRTC illustrating areas with higher latency along the test route. Full quantitative results for all protocols will be presented in the final paper.

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